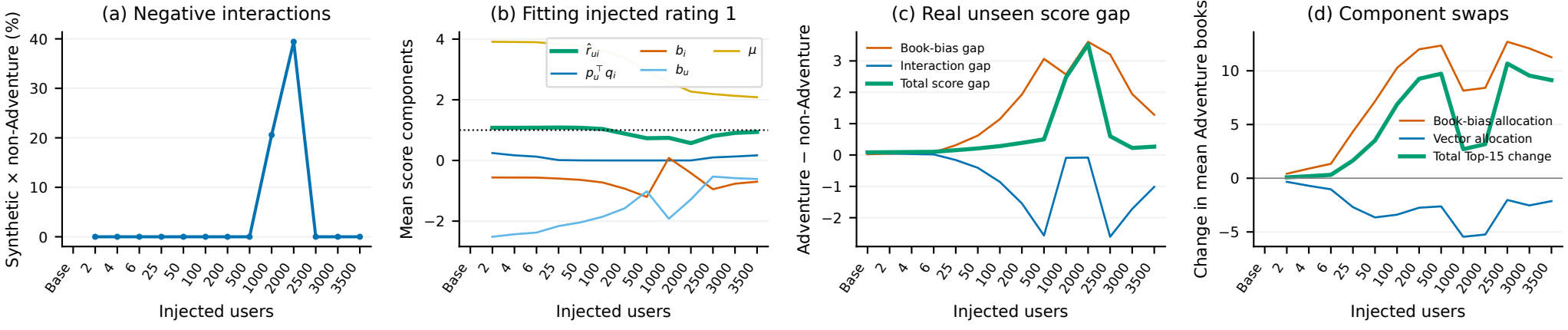
NMF | Adventure injection
(a) Negative interactions
Synthetic × non-Adventure (%)
Injected users
(b) Fitting injected rating 1
Mean score components
$\hat{r}_{ui}$
$p_u^\top q_i$
b_i
b_u
μ
Injected users
(c) Real unseen score gap
Adventure − non-Adventure
Book-bias gap
Interaction gap
Total score gap
Injected users
(d) Component swaps
Change in mean Adventure books
Book-bias allocation
Vector allocation
Total Top-15 change
Injected users
Base
2
4
6
25
50
100
200
500
1000
2000
2500
3000
3500
NMF factor sign check failed at n = 1000, 2000, 2500. Raw results retained.